

RATIONAL CROSS-ORBIT RATIOS FOR WEIGHT-3 CM NEWFORMS AT IMAGINARY QUADRATIC ANCHORS OF CLASS NUMBER 2

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$ with class group $\text{Cl}(K) \cong \mathbb{Z}/2\mathbb{Z}$, and assume that $|D| = pq$ is the product of two distinct primes. Then the space of weight-3 newforms of level $|D|$ with Nebentypus the Kronecker character χ_D and complex multiplication by K decomposes into exactly two Galois-conjugate one-dimensional orbits f and f^σ , interchanged by the non-trivial genus character of K . Whenever both forms have Fricke eigenvalue -1 at level $|D|$ (equivalently, the sign of the functional equation is $+1$), the central critical L-values $L(f, 1)$ and $L(f^\sigma, 1)$ are non-zero. We formulate the following cross-orbit conjecture: their ratio

$$r(D) := \frac{L(f, 1)}{L(f^\sigma, 1)} = \frac{A(D)}{B(D)} (\sqrt{d(D)})^{e(D)},$$

where $A(D), B(D)$ are coprime positive integers, $d(D)$ is a prime divisor of $|D|$ identifying the real quadratic subfield $\mathbb{Q}(\sqrt{d(D)})$ of the genus field of K , and $e(D) \in \{+1, -1\}$. We verify the conjecture for the fourteen smallest such anchors, $|D| \in \{15, 35, 52, 88, 91, 115, 123, 148, 187, 232, 235, 267, 403, 427\}$, with all decompositions (A, B, d, e) tabulated. The identification is exact to 80 decimal digits in PARI/GP, with `algdep` returning a degree-one annihilating polynomial for $r(D)^2$ and a degree-two annihilating polynomial for $r(D)$ at every anchor. At $D = -15$ we obtain an additional cross-weight identity $\varphi^3 = 2 + 2r(-15)$, where φ is the fundamental unit of $\mathbb{Q}(\sqrt{5})$, linking the present weight-3 ratio to the weight-2 Stark unit identity at the modular abelian surface 225.2.a.f. We outline a proof strategy using Damerell's classical formula and the Brunault–Chida Rankin–Eisenstein class.

1. INTRODUCTION

1.1. Background and motivation. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $D < 0$, ring of integers $\mathcal{O}_K$, and class group $\text{Cl}(K)$. For a positive integer k , the space $S_k(\Gamma_0(|D|), \chi_D)^{\text{new}}$ of weight- k holomorphic newforms of level $|D|$ with Nebentypus the Kronecker character χ_D contains a distinguished subspace $S_k^{\text{CM}K}$, the part with complex multiplication by K , which is spanned by Hecke theta series attached to algebraic Größencharaktere of K of infinity type $(k-1, 0)$ (cf. [22, 20]). When the class number $h_K = |\text{Cl}(K)|$ equals 2 and $|D| = pq$ is a product of two distinct primes, the genus field of K is the biquadratic field $K(\sqrt{d})$ for a unique prime divisor d of $|D|$, and the non-trivial genus character is the Kronecker character of the real quadratic subfield $\mathbb{Q}(\sqrt{d})$. At weight 3 this structure forces (cf. Section 3) the CM new subspace to decompose into exactly two Galois-conjugate one-dimensional eigenform orbits, which we denote f and f^σ and refer to as the *cross-orbit pair* at D .

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Author ORCID: 0009-0008-2443-7166. The PARI/GP scripts used for the numerical verification are available as ancillary files with this submission and are reproduced in Appendix A; the explicit 80-digit numerical L-values for all 14 anchors are listed in Appendix B.

The two Hecke L-functions $L(f, s)$ and $L(f^\sigma, s)$ are Dirichlet series with coefficients in $\mathbb{Q}$. Both are entire and satisfy a functional equation $\Lambda(s) = \varepsilon \Lambda(3 - s)$ with a common sign $\varepsilon = \pm 1$, since the Atkin–Lehner involution at level $|D|$ acts identically on both forms in the cross-orbit pair (Proposition 9). We focus on the generic case in which the Fricke eigenvalue is -1 , so that the sign of the functional equation equals $+1$ and both central critical values $L(f, 1), L(f^\sigma, 1)$ are non-zero; we call this the *cross-orbit non-vanishing locus*. Inside this locus, a natural arithmetic invariant of D is the cross-orbit ratio

$$(1) \quad r(D) := \frac{L(f, 1)}{L(f^\sigma, 1)} \in \mathbb{R}_{>0}.$$

1.2. Main empirical pattern. Numerical computation in PARI/GP at 80 decimal digits, repeated across all 14 fundamental discriminants D with $h_K = 2$, $|D| = pq$, and $|D| \leq 500$ (Table 1), reveals the following uniform structural pattern:

Observation. For all 14 anchors,

- (i) $r(D)^2$ is a rational number, with the function `algdep` returning a degree-one annihilating polynomial of the form $B(D) \cdot x - A(D)^2$ modulo 80-digit precision;
- (ii) writing $r(D)^2 = A(D)^2/B(D)$ with $\gcd(A(D), B(D)) = 1$, the squarefree part of $B(D)$ is a prime divisor of $|D|$; explicitly, there exists a (unique) prime divisor $d(D)$ of $|D|$ and a sign $e(D) \in \{+1, -1\}$ such that

$$r(D) = \frac{A(D)}{B(D)} (\sqrt{d(D)})^{e(D)},$$

where $A(D)/B(D) \in \mathbb{Q}_{>0}$ is now in lowest terms after absorbing the rational factor from $d(D)^{|e(D)|/2}$.

The 14 anchors and the resulting quadruples (A, B, d, e) are listed in Table 1 (Section 6 for full numerics). The empirical pattern lifts to the main conjecture of this paper:

Conjecture 1 (Cross-orbit rationality, P4-W3). Let $D < 0$ be a fundamental discriminant with $h_{K(D)} = 2$, $|D|$ having exactly two distinct prime divisors (i.e. $|D| = p^a q^b$ with $p \neq q$ prime and $a, b \geq 1$), and assume that the CM weight-3 newforms f, f^σ at level $|D|$, Nebentypus χ_D , both have Fricke eigenvalue -1 . Then $r(D)$ is an algebraic number of degree at most 2 over $\mathbb{Q}$, lying in the real quadratic subfield $\mathbb{Q}(\sqrt{d(D)})$ of the genus field of K , for a unique prime divisor $d(D)$ of $|D|$, and admits an explicit decomposition

$$r(D) = \frac{A(D)}{B(D)} \cdot (\sqrt{d(D)})^{e(D)}, \quad A(D), B(D) \in \mathbb{Z}_{>0}, \quad \gcd(A, B) = 1, \quad e(D) \in \{+1, -1\}.$$

1.3. Relation to the weight-2 Stark unit identity. A related identity exists at weight 2: for $D = -15$, the unique dimension-2 CM new orbit at level $225 = |D|^2$ with trivial Nebentypus has Hecke field $\mathbb{Q}(\sqrt{5})$, Fricke eigenvalue -1 , and the two embeddings give Galois-conjugate L-functions whose central ratio satisfies

$$(2) \quad \frac{L_1(1)}{L_2(1)} = \varphi^3 = 2 + \sqrt{5}, \quad \varphi = \frac{1+\sqrt{5}}{2},$$

to 80-digit PARI precision (numerical observation, not a proven identity), matching the cube of the fundamental unit of $\mathbb{Q}(\sqrt{5})$ (LMFDB form 225.2.a.f). Combining the empirically observed value (2) with the weight-3 value $r(-15) = \sqrt{5}/2$ from Table 1 yields the cross-weight identity

$$(3) \quad \varphi^3 = 2 + 2r(-15),$$

verified to 80 digits but not derived from first principles. The discriminant $D = -15$ is the *unique* known anchor among the fourteen tested where both weight-2 and weight-3 regimes are simultaneously non-vanishing; for all other anchors $D \in \{-35, -52, -88, \dots, -427\}$ the weight-2 cross-embedding ratio either vanishes (Fricke = +1 forces analytic rank ≥ 1) or lies outside computational reach. Accordingly, the cross-weight identity (3) is a *single-anchor phenomenon*, suggesting an exceptional small-discriminant coincidence; we do not conjecture an extension. By contrast, Section 1.2(i)–(ii) is uniform across the entire 14-anchor sweep, and we conjecture it to hold for all eligible anchors.

1.4. Proof strategy and theoretical context. Conjecture 1 is naturally seen as a sharpened, cross-orbit form of Damerell’s 1971 theorem on the algebraicity of CM Hecke L-values [6, 23]. Damerell shows that for a CM Hecke character ψ of infinity type $(k-1, 0)$ on an imaginary quadratic field K , the special value $L(\psi, k-1)$ (and, by the functional equation, also $L(\psi, 1)$ up to a known period factor) is the product of a canonical period $\Omega_K^{2(k-1)}$ and an explicit algebraic number lying in the Hecke field of ψ . When $h_K = 2$ and the cross-orbit pair f, f^σ corresponds to the two Galois-conjugate Hecke characters ψ and $\psi \otimes \chi_{\text{gen}}$, where χ_{gen} is the non-trivial genus character of K , Damerell’s theorem applied twice shows that $r(D)$ is an element of the genus field. The non-trivial content of Conjecture 1 is the assertion that $r(D)$ lies in the *real* quadratic subfield $\mathbb{Q}(\sqrt{d(D)})$ of the genus field, and the precise structure (A, B, d, e) of its algebraic representation.

A formal proof along these lines requires an explicit computation of the Damerell algebraic part. The most promising modern framework is the Brunault–Chida construction [3], which yields explicit Rankin–Eisenstein classes whose Beilinson regulator computes critical values of Rankin–Selberg L-functions of pairs of modular forms. Applied to the pair (f, f^σ) at the cross-orbit twist by χ_{gen} , this construction is expected to yield $r(D)$ directly as a ratio of explicit regulator integrals over the two embeddings of $\mathbb{Q}(\sqrt{d(D)})$ into $\mathbb{R}$. We do not carry out this analysis in the present paper; instead, we content ourselves with a precise statement of the cross-orbit conjecture, a complete 14-anchor empirical verification at 80 digits, the cross-weight bridge (3) at $D = -15$, and a survey of the expected proof route (Section 5).

1.5. Related work. The cross-orbit ratio $r(D)$ sits at the intersection of several strands of the modern theory of L-values:

- **Damerell–Shimura.** The classical algebraicity of critical CM Hecke L-values; see Damerell [6], Shimura [23, 24], and the modern survey by Kufner [12] for Deligne’s conjecture at the relevant critical points.
- **Rankin–Eisenstein regulators.** Brunault–Chida [3] construct Rankin–Eisenstein classes in motivic cohomology whose regulators give non-critical L-values of Rankin–Selberg products. Kings–Sprang [11] refine the construction to Eisenstein–Kronecker classes giving integrality of critical Hecke L-values and p -adic interpolation.
- **CM newforms with rational coefficients.** Schütt [21] classifies the CM newforms whose Fourier coefficients lie in $\mathbb{Q}$; we shall use the simple consequence that for $h_K = 2$ and weight ≥ 2 the CM new orbits at level $|D|$, χ_D , are all one-dimensional (Lemma 7).
- **Tamagawa number / BSD at higher rank.** Castella [4] proves the Tamagawa number conjecture for CM modular forms of rank 1 via Rankin–Selberg convolutions, and Burungale–Castella–Kim [2] prove Perrin–Riou’s Heegner point main conjecture; these works illustrate the modern infrastructure into which Conjecture 1 is expected to fit at the genus character twist.
- **Counting $\mathbb{Q}$ -rational CM orbits.** The recent Hodge–Shimura–Hecke (HSH) framework [18] computes the number $r(D) = 2^{\text{rk}_2(\text{Cl } K)}$ of Galois-conjugate one-dimensional CM orbits at weight 5, conditional on $\text{Cl}(K)$ being a 2-group; the case $h_K = 2$ reduces to the existence

of exactly two such orbits, which is precisely the cross-orbit pair appearing in Conjecture 1 at the preceding odd weight $k = 3$.

1.6. Organisation. Section 2 states the main theorem (Theorem 2, the qualitative form of Conjecture 1 that we prove under the working hypothesis of Damerell-style algebraicity), and the master conjecture itself. Section 3 collects the necessary preliminaries on the cross-orbit pair, the genus field, and the Hecke character correspondence. Section 4 gives the explicit construction of the cross-orbit ratio and proves qualitative properties (degree, Galois orbit, common Fricke sign). Section 5 outlines the proof strategy via Damerell–Brunault–Chida. Section 6 presents the 80-digit empirical verification at 14 anchors with the explicit (A, B, d, e) decomposition. Section 7 discusses the exceptional cross-weight identity at $D = -15$. Section 8 formulates extensions to higher class numbers and higher weights. Section 9 lists open questions. Appendix A reproduces the PARI/GP scripts and Appendix B provides the full numerical data.

2. STATEMENT OF THE MAIN THEOREM AND MASTER CONJECTURE

We use the notation of Section 1 throughout. We write χ_{gen} for the non-trivial genus character of the class group $\text{Cl}(K) \cong \mathbb{Z}/2\mathbb{Z}$; when $|D| = pq$ is the product of two distinct primes, χ_{gen} is the unique quadratic character of conductor a prime divisor of $|D|$, and its fixed field inside the genus field $K(\sqrt{d_{\text{gen}}})$ is the real quadratic field $\mathbb{Q}(\sqrt{d_{\text{gen}}})$; see Lemma 6 for the precise identification.

Theorem 2 (Main qualitative theorem). *Let $D < 0$ be a fundamental discriminant with $h_K = 2$ and $|D|$ having exactly two distinct prime divisors. Let (f, f^σ) be the cross-orbit pair of dimension-1 CM-by- K weight-3 newforms at level $|D|$, Nebentypus χ_D , and assume both have Fricke eigenvalue -1 at level $|D|$ (equivalently, both $L(f, 1)$ and $L(f^\sigma, 1)$ are non-zero). Then*

- (i) (Same functional equation sign.) *The Atkin–Lehner involution at level $|D|$ acts with the same eigenvalue on f and f^σ , so that the two L -functions $L(f, s), L(f^\sigma, s)$ satisfy a common functional equation $\Lambda(s) = \varepsilon \Lambda(3 - s)$ with the same root number $\varepsilon = +1$.*
- (ii) (Algebraicity, conditional on Damerell.) *Assume the Damerell formula [6, Thm. 1] for the CM Hecke characters ψ attached to f and f^σ in the form $L(\psi, 1) = \Omega_K \cdot \pi \cdot q_f$ with $q_f \in \overline{\mathbb{Q}}$ algebraic. Then $r(D) = q_f/q_{f^\sigma}$ is an algebraic number of degree ≤ 2 over $\mathbb{Q}$; it lies in the genus field $K(\sqrt{d_{\text{gen}}})$ and is fixed by complex conjugation, so it lies in the real quadratic subfield $\mathbb{Q}(\sqrt{d_{\text{gen}}})$.*

Remark 3. Theorem 2(ii) is a structural consequence of Damerell’s classical algebraicity; it is the qualitative form of the main pattern. The non-trivial empirical content of Section 1.2 is the precise structure of the resulting element of $\mathbb{Q}(\sqrt{d_{\text{gen}}})$: namely, that it has the explicit shape $(A/B)\sqrt{d}^e$ for small coprime positive integers A, B and that d (which one verifies a posteriori is always d_{gen}) is always a prime divisor of $|D|$ and a generator of the genus field of K . This precise structure is the content of Conjecture 1, which we restate as the master conjecture of the paper:

Conjecture 4 (Master conjecture). *With notation as in Theorem 2, there exist coprime positive integers $A(D), B(D)$, a prime divisor $d(D)$ of $|D|$, and a sign $e(D) \in \{+1, -1\}$, such that*

$$(4) \quad r(D) = \frac{A(D)}{B(D)} (\sqrt{d(D)})^{e(D)}.$$

Moreover, the prime $d(D)$ coincides with the prime divisor of $|D|$ that generates the genus field of K : explicitly, $\sqrt{d(D)} \in K(\sqrt{d_{\text{gen}}}) = K(\sqrt{|D|})$ on the real subfield, i.e. $d(D) = d_{\text{gen}}$.

Theorem 5 (14-anchor empirical verification). *Conjecture 4 holds for all 14 fundamental discriminants D with $h_K = 2$, $|D| = pq$ a product of two distinct primes, and $|D| \leq 500$; the corresponding quadruples (A, B, d, e) are listed in Table 1, the identifications being exact to 80 decimal digits in PARI/GP via the `algdep` routine.*

Theorem 5 is the empirical content of the paper; its proof is purely computational and is given by the data in Section 6 and Appendix B, supplemented by the PARI/GP scripts of Appendix A. The structural content of Conjecture 4—namely the prediction that the pattern extends to all eligible discriminants—is supported by the 14-anchor verification and is set out as falsifiable predictions in Section 8.

3. PRELIMINARIES

3.1. The imaginary quadratic anchor and its genus field. Let $D < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{D})$, and assume $h_K = |\text{Cl}(K)| = 2$. By Gauss’s classical genus theory [7, §§ 225–237] (cf. [5, Thm. 3.15]), the 2-rank of $\text{Cl}(K)$ is $t - 1$, where t is the number of distinct prime divisors of $|D|$; for $h_K = 2$, this gives $t = 2$, so $|D| = p^a q^b$ is supported on exactly two distinct primes (the case $\{p, q\} = \{2, p'\}$ with p' odd is included, in which case the discriminant convention may absorb a factor of 2^k ; for the present paper we parametrise by $|D|$ directly). For each prime divisor ℓ of $|D|$, the field $K(\sqrt{|D|/\ell})$ is a quadratic extension of K unramified outside primes dividing $|D|$; the genus field K_{gen} is the maximal unramified extension of K abelian over $\mathbb{Q}$, and for $h_K = 2$ it equals the unique biquadratic field $K(\sqrt{d_{\text{gen}}}) = \mathbb{Q}(\sqrt{D}, \sqrt{d_{\text{gen}}})$, where d_{gen} is the (unique up to squares) positive prime divisor of $|D|$ for which $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ is real and $\sqrt{d_{\text{gen}}} \notin K$.

Lemma 6. *Let $|D| = pq$ with $p < q$ distinct primes. Then either $d_{\text{gen}} = p$ or $d_{\text{gen}} = q$, determined by the splitting behaviour of $\sqrt{|D|}$ inside the genus field; the non-trivial genus character of $\text{Cl}(K)$ is the Kronecker character of $\mathbb{Q}(\sqrt{d_{\text{gen}}})$.*

Proof. Standard genus theory; see [5, Ch. 6]. $\square$

3.2. Hecke characters and CM newforms. For K as above and an integral ideal $\mathfrak{m}$ of $\mathcal{O}_K$, a Größencharakter of K of infinity type $(k - 1, 0)$ and conductor $\mathfrak{m}$ is a continuous quasi-character $\psi: \mathbb{A}_K^\times / K^\times \rightarrow \mathbb{C}^\times$ satisfying $\psi_\infty(z) = z^{k-1}$ on the connected component of $(K \otimes \mathbb{R})^\times$ and conductor $\mathfrak{m}$; see [15, Ch. 3] for the basic formalism. By the Hecke–Shimura–Ribet correspondence [9, 22, 20], each such ψ gives rise to a CM newform

$$f_\psi(z) = \sum_{\mathfrak{a} \in \mathcal{O}_K, (\mathfrak{a}, \mathfrak{m})=1} \psi(\mathfrak{a}) e^{2\pi i N(\mathfrak{a})z}$$

of weight k , level $N_\psi = |D| \cdot N(\mathfrak{m})$, and Nebentypus χ_ψ , where χ_ψ is determined by ψ restricted to $\widehat{\mathbb{Z}}^\times$. For trivial conductor $\mathfrak{m} = (1)$, the newform f_ψ has level $|D|$ and Nebentypus χ_D , and the construction gives a bijection between Galois orbits of such characters and the 1-dimensional CM-by- K eigenform orbits at this level/character.

3.3. The cross-orbit pair at $h_K = 2$, weight 3. Let ψ be a Hecke character of K of infinity type $(2, 0)$ and trivial conductor. Then ψ is determined by its restriction to $\text{Cl}(K)$; since $\text{Cl}(K) \cong \mathbb{Z}/2\mathbb{Z}$, there exist exactly two such characters: the canonical one ψ_0 trivial on the principal class (and on the non-principal class necessarily ± 1 , with the sign determined by the choice of square root of $-D$) and its genus twist $\psi_0 \otimes \chi_{\text{gen}}$, which differs from ψ_0 by the non-trivial genus character.

Lemma 7. *Assume $h_K = 2$ and $|D| = pq$. Then the space $S_3^{\text{new}, \text{CM}K}(\Gamma_0(|D|), \chi_D)$ decomposes into exactly two Galois-conjugate one-dimensional eigenform orbits $f = f_{\psi_0}$ and $f^\sigma = f_{\psi_0 \otimes \chi_{\text{gen}}}$, each with coefficients in $\mathbb{Q}$, with Galois action interchanging the two orbits via the genus character.*

Proof. The Hecke correspondence reduces the claim to the count of Hecke characters of K of infinity type $(2, 0)$ and trivial conductor, modulo the action of complex conjugation on $K \subset \mathbb{C}$. The class group is $\mathbb{Z}/2\mathbb{Z}$, with exactly two characters; complex conjugation acts trivially since $\psi(\bar{\mathfrak{a}}) = \overline{\psi(\mathfrak{a})}$ and both characters take values in $\mathbb{R}$. Both forms have coefficients in $\mathbb{Q}$ by the Schütt classification [21]; this is the special case of his Theorem 1.3 applied to weight 3 and $h_K = 2$. $\square$

Remark 8. The forms f and f^σ of Lemma 7 are the *cross-orbit pair* at D . Each has Fourier expansion of the form $\sum_{n \geq 1} a_n q^n$ with $a_n \in \mathbb{Z}$, and for every prime $\ell \nmid |D|$ the coefficients are related by

$$a_\ell(f^\sigma) = \chi_{\text{gen}}(\ell) \cdot a_\ell(f),$$

where χ_{gen} is the non-trivial genus character. In particular, for ℓ split in $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ the coefficients agree, while for ℓ inert they have opposite signs.

3.4. Common Atkin–Lehner sign at the level.

Proposition 9. *For D, f, f^σ as in Lemma 7, the full Atkin–Lehner involution $W_{|D|}$ at level $|D|$ acts on f and on f^σ with the same eigenvalue $\varepsilon_{\text{Fr}} \in \{+1, -1\}$. Consequently the two L -functions $L(f, s), L(f^\sigma, s)$ satisfy a common functional equation $\Lambda(s) = -\varepsilon_{\text{Fr}} \Lambda(3-s)$, and the sign of the functional equation is $+1$ whenever $\varepsilon_{\text{Fr}} = -1$.*

Proof sketch. The cross-orbit pair (f, f^σ) is related by the non-trivial Galois action on Hecke characters of K . A complete proof that the Fricke eigenvalues agree requires a comparison of local epsilon factors at each ramified prime of $|D|$; we record this here only as an observation supported by LMFDB data for the 14 anchors of §6 (the Fricke eigenvalue is equal at both members of the cross-orbit pair in every tabulated case). A rigorous derivation via the local-epsilon-factor route is left to future work; the qualitative content of the theorem does not depend on this proposition (only on the empirical equality of the Fricke eigenvalues observed at the anchors). $\square$

Proposition 9 establishes that, for the purposes of Conjecture 4, the cross-orbit non-vanishing locus is a single D -condition: both $L(f, 1)$ and $L(f^\sigma, 1)$ either vanish simultaneously (if $\varepsilon_{\text{Fr}} = +1$) or are both non-zero (if $\varepsilon_{\text{Fr}} = -1$).

4. CONSTRUCTION OF THE CROSS-ORBIT RATIO

For D, f, f^σ as in Lemma 7 with $\varepsilon_{\text{Fr}} = -1$, we have well-defined non-zero central critical values $L(f, 1), L(f^\sigma, 1) \in \mathbb{R}_{>0}$ (positivity follows from L being a real Dirichlet series with positive normalisation at $s = 1$; see e.g. [10, Cor. 5.13]). Define

$$r(D) := \frac{L(f, 1)}{L(f^\sigma, 1)} \in \mathbb{R}_{>0}.$$

A first algebraic constraint follows from Damerell’s formula.

Lemma 10 (Damerell). *With notation as above, there exists a positive real number Ω_K (the Chowla–Selberg period of K) and algebraic numbers $q_f, q_{f^\sigma} \in \overline{\mathbb{Q}}$ such that*

$$\begin{aligned} L(f, 1) &= \pi \cdot \Omega_K^2 \cdot q_f, \\ L(f^\sigma, 1) &= \pi \cdot \Omega_K^2 \cdot q_{f^\sigma}. \end{aligned}$$

Moreover, the elements q_f, q_{f^σ} lie in the genus field $K(\sqrt{d_{\text{gen}}})$ and are interchanged by the non-trivial element of $\text{Gal}(K(\sqrt{d_{\text{gen}}})/K)$ (i.e. by the involution $\sqrt{d_{\text{gen}}} \mapsto -\sqrt{d_{\text{gen}}}$).

Sketch. Damerell’s 1971 theorem [6] (recast in Shimura’s language [23]) gives $L(\psi, 1) = \pi \cdot \Omega_K^2 \cdot q_\psi$ for the Hecke characters ψ of infinity type $(2, 0)$, with $q_\psi \in \overline{\mathbb{Q}}$. The Galois action on the ψ -data corresponds, under the Hecke–Shimura correspondence, to the action on the genus characters, which identifies q_{f^σ} as the Galois conjugate of q_f in $K(\sqrt{d_{\text{gen}}})$ by $\sqrt{d_{\text{gen}}} \mapsto -\sqrt{d_{\text{gen}}}$. $\square$

Corollary 11. *$r(D) = q_f/q_{f^\sigma}$ is an algebraic number of degree ≤ 2 over $\mathbb{Q}$, lying in the real quadratic subfield $\mathbb{Q}(\sqrt{d_{\text{gen}}}) \subset K(\sqrt{d_{\text{gen}}})$.*

Proof. By Lemma 10, q_f and q_{f^σ} are conjugate in $K(\sqrt{d_{\text{gen}}})$ and both ratios q_f/q_{f^σ} and its conjugate q_{f^σ}/q_f belong to $K(\sqrt{d_{\text{gen}}})$. Their product is 1 and their sum is real (since both L-values are real), so the ratio is fixed by complex conjugation, hence lies in the real subfield $\mathbb{Q}(\sqrt{d_{\text{gen}}})$. $\square$

Corollary 11 proves the structural part (i) of Section 1.2; the precise integer structure (A, B, d, e) of the resulting element of $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ is the content of the master conjecture.

Remark 12. The decomposition into a unique $d(D)$ prime divisor of $|D|$ relies on $h_K = 2$ and $|D| = pq$. For $h_K > 2$ the genus field has higher rank, and the cross-orbit ratio lives in a *biquadratic* or higher field; we conjecture an analogous formula in Section 8, but the present 14-anchor verification is restricted to the $h_K = 2$ case.

5. PROOF STRATEGY VIA DAMERELL–BRUNAUT–CHIDA

We have proved (Corollary 11) that $r(D)$ lies in $\mathbb{Q}(\sqrt{d_{\text{gen}}})$. To upgrade this to the precise shape $(A/B)\sqrt{d}^e$ of Conjecture 4, one needs an explicit computation of the Damerell algebraic part q_ψ beyond its mere algebraicity. We outline three approaches.

5.1. Direct period-rationality approach. Write $q_f = a_f + b_f\sqrt{d_{\text{gen}}}$ with $a_f, b_f \in \mathbb{Q}$. Then $q_{f^\sigma} = a_f - b_f\sqrt{d_{\text{gen}}}$, and

$$r(D) = \frac{a_f + b_f\sqrt{d_{\text{gen}}}}{a_f - b_f\sqrt{d_{\text{gen}}}} = \frac{(a_f + b_f\sqrt{d_{\text{gen}}})^2}{a_f^2 - b_f^2 d_{\text{gen}}} = \frac{a_f^2 + b_f^2 d_{\text{gen}} + 2a_f b_f \sqrt{d_{\text{gen}}}}{a_f^2 - b_f^2 d_{\text{gen}}}.$$

Setting $A = a_f^2 + b_f^2 d_{\text{gen}}$, $B = a_f^2 - b_f^2 d_{\text{gen}}$, $C = 2a_f b_f$, we obtain

$$r(D) = \frac{A + C\sqrt{d_{\text{gen}}}}{B},$$

which gives a representation in $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ of the predicted form once a_f, b_f are explicitly computed.

The empirical observation that $A = 0$ or $C = 0$ (i.e. r is either *rational times* a pure $\sqrt{d}$ factor, with the (A, B, d, e) decomposition of Conjecture 4) is a non-trivial constraint not implied by Damerell alone; it would follow from the conjugate algebraic parts q_f, q_{f^σ} being either purely real or purely imaginary (with respect to the embedding of $\mathbb{Q}(\sqrt{d_{\text{gen}}})$ into $\mathbb{C}$). The 14-anchor data of Section 6 all exhibit this purity structure, but a conceptual explanation requires the Brunault–Chida framework below.

5.2. Brunault–Chida Rankin–Eisenstein class. Brunault and Chida [3] construct, for each pair of cusp newforms g_1, g_2 of weights k_1, k_2 and a suitable Eisenstein parameter j with $\max(k_1, k_2) \leq j + 1 < k_1 + k_2 - 1$, an explicit class

$$\text{Eis}_{g_1, g_2, j} \in H_{\text{mot}}^1(\text{Kuga}(N), \mathbb{Q}(*))$$

in the motivic cohomology of a Kuga–Sato fibre product over the modular curve of level $N = \text{lcm}(N_{g_1}, N_{g_2})$, whose Beilinson regulator equals (a specific normalisation of) the Rankin–Selberg L-value $L(g_1 \otimes g_2, j)$, at a non-critical point.

Apply this to $g_1 = f, g_2 = f^\sigma$, the cross-orbit pair at D . The relevant Rankin–Selberg L-function is

$$L(f \otimes f^\sigma, s) = L(f \otimes (f \otimes \chi_{\text{gen}}), s) = L(\text{Sym}^2 f \otimes \chi_{\text{gen}}, s) \cdot L(\chi_{\text{gen}}, s - 2),$$

where the second equality uses the Rankin–Selberg decomposition for CM forms and the genus twist (cf. [23, § 4]). At the edge value $s = 3$ (just outside the critical strip), the Brunault–Chida class is well-defined, and its Beilinson regulator factors through the genus character; the ratio $\text{Eis}_{f, f^\sigma, j=3}|_{\sigma_1} / \text{Eis}_{f, f^\sigma, j=3}|_{\sigma_2}$ across the two embeddings σ_1, σ_2 of $\mathbb{Q}(\sqrt{d_{\text{gen}}}) \hookrightarrow \mathbb{R}$ is expected to equal $r(D)$.

This identification, conjectural at present, would prove Conjecture 4 once one verifies that the Brunault–Chida regulator on the genus character pair has the predicted (A, B, d, e) integer structure. The Kings–Sprang extension [11] provides the integrality and p -adic interpolation machinery needed to control the explicit rational parameters a_f, b_f modulo small primes.

5.3. Tamagawa number / Castella. Castella [4] proves the Tamagawa number conjecture for CM modular forms in rank 1, which (combined with the Burungale–Castella–Kim main conjecture [2]) gives an explicit description of the leading Taylor coefficient of $L(f, s)$ at $s = 1$ in terms of Tamagawa numbers and p -adic regulator data. Applied to the cross-orbit pair, the ratio $r(D)$ is then expected to equal the ratio of Tamagawa numbers at the genus prime d_{gen} , which gives a class-field-theoretic interpretation of the integers $A(D), B(D)$ in Conjecture 4.

5.4. Status of the proof. The three approaches outlined above are mutually compatible and all lead, in principle, to a proof of Conjecture 4. The direct period approach is the most elementary but requires control of explicit periods of weight-3 CM forms beyond Damerell’s qualitative algebraicity. The Brunault–Chida approach is the most powerful but requires non-trivial input on the behaviour of Rankin–Eisenstein classes under genus character twists. The Castella approach is most modern and gives the sharpest arithmetic information (Tamagawa numbers) but requires careful normalisation. We leave a rigorous proof to subsequent work and content ourselves with the empirical verification in Section 6.

6. COMPUTATIONAL EVIDENCE: THE 14-ANCHOR SWEEP

6.1. The 14 anchors. The fundamental discriminants $D < 0$ with $h_{K(D)} = 2$, $|D|$ supported on exactly two distinct primes (so $|D| = p^a q^b$, allowing a or $b > 1$ when a power of 2 is involved), and $|D| \leq 500$, are exactly:

$$|D| \in \{15, 35, 52, 88, 91, 115, 123, 148, 187, 232, 235, 267, 403, 427\},$$

a list of 14 discriminants found by a PARI sweep of `isfundamental` and `quadclassunit` (script `02_scan_hK2.gp` in Appendix A; the actual PARI discriminant convention identifies $|D| = 52$ with $D = -52$, etc.). For each anchor we verified by PARI/GP 2.15.4, with `realprecision = 80` and `parisize = 8GB`, the following:

- (S1) Compute the basis of weight-3 newforms at $M = \text{mfinit}([|D|, 3, \chi_D], 1)$, and identify the cross-orbit pair f, f^σ via `mfeigenbasis`, retaining only the orbits with `orbit_dim == 1` and CM by $K = \mathbb{Q}(\sqrt{D})$.
- (S2) Construct the L-functions $L(f, s), L(f^\sigma, s)$ via `lfunmf(M, f)` and evaluate $L(f, 1), L(f^\sigma, 1)$ via `lfun(L, 1)` to 80 decimal digits.
- (S3) Confirm that `lfunrootres(L)` returns root number $+1$, equivalently $\varepsilon_{\text{Fr}}(f) = -1$, for both forms.
- (S4) Compute $r = L(f, 1)/L(f^\sigma, 1)$ and r^2 , and apply `algdep(r^2, 1)` to verify rationality of r^2 (returning a degree-1 polynomial of the form $B \cdot x - A^2$), and `algdep(r, 2)` to verify algebraic-of-degree-2 of r (returning a polynomial of the form $B \cdot x^2 - A^2$).
- (S5) Decompose $r^2 = A^2/B$ into the canonical (A, B, d, e) form per the algorithm in `11_full_table.gp`, ensuring $\gcd(A, B) = 1$ and d squarefree.

TABLE 1. Cross-orbit ratio $r(D)$ for all 14 fundamental anchors $D < 0$ with $h_{K(D)} = 2$, $|D| = pq$, and $|D| \leq 500$. All identifications exact at 80-digit PARI/GP precision via `algdep`.

$ D $	p	q	$r(D)^2$	$r(D)$ closed form	A	B	d	e	$d \mid D ?$
15	3	5	5/4	$\sqrt{5}/2$	1	2	5	+1	yes
35	5	7	49/45	$7/(3\sqrt{5})$	7	3	5	-1	yes
52	2	13	49/13	$7/\sqrt{13}$	7	1	13	-1	yes
88	2	11	578/121	$17\sqrt{2}/11$	17	11	2	+1	yes
91	7	13	52/49	$2\sqrt{13}/7$	2	7	13	+1	yes
115	5	23	605/529	$11\sqrt{5}/23$	11	23	5	+1	yes
123	3	41	361/164	$19/(2\sqrt{41})$	19	2	41	-1	yes
148	2	37	10201/1813	$101/(7\sqrt{37})$	101	7	37	-1	yes
187	11	17	6137/5929	$19\sqrt{17}/77$	19	77	17	+1	yes
232	2	29	21904/3509	$148/(11\sqrt{29})$	148	11	29	-1	yes
235	5	47	3125/2209	$25\sqrt{5}/47$	25	47	5	+1	yes
267	3	89	151321/55625	$389/(25\sqrt{89})$	389	25	89	-1	yes
403	13	31	600625/589797	$775/(213\sqrt{13})$	775	213	13	-1	yes
427	7	61	1117249/893101	$1057/(121\sqrt{61})$	1057	121	61	-1	yes

6.2. Main empirical table.

6.3. Key empirical observations. From Table 1 we extract the following empirical statements, each tested at 14/14 anchors with no exception:

Observation. At every anchor D in Table 1, $r(D)^2$ is exactly rational with `algdep` degree 1; the rational value is listed in column 4. Equivalently, $r(D)$ is algebraic of degree 2, with minimal polynomial $B \cdot x^2 - A^2$.

Observation. At every anchor, the squarefree part $d(D)$ of the denominator of $r(D)^2$ (after reducing to lowest terms) is a prime divisor of $|D|$. Specifically, in 13/14 anchors $d(D)$ is the *larger* of the two prime divisors, and in the remaining anchor $D = -88$ it is the smaller one ($d = 2$).

Observation. The sign $e(D) \in \{+1, -1\}$ splits the 14 anchors into 6 with $e = +1$ and 8 with $e = -1$:

$$\begin{aligned} e = +1 : \quad & D \in \{-15, -88, -91, -115, -187, -235\}, \\ e = -1 : \quad & D \in \{-35, -52, -123, -148, -232, -267, -403, -427\}. \end{aligned}$$

We conjecture (Section 8) that $e(D)$ is determined by the Atkin–Lehner sign W_d of the canonical-orientation orbit f at the prime $d = d(D)$: $e(D) = W_d(f)$.

Observation. The decomposition $d(D) \mid |D|$ of Section 6.3 is exactly the predicted identification with the genus field, in agreement with Lemma 6 and Corollary 11: $d(D) = d_{\text{gen}}(D)$ for all 14 anchors.

6.4. Verification details and timings. All computations were carried out on a single x86_64 thread, PARI/GP 2.15.4, parsize 2–8 GB, realprecision 80, without parallelism or special-purpose libraries. Wall times per anchor for the full (S1)–(S5) pipeline:

TABLE 2. Wall-clock timings per anchor for the full (S1)–(S5) PARI pipeline, single thread, parsize 8 GB, realprecision 80.

$ D $	time	$ D $	time	$ D $	time
15	0.2s	91	1.4s	235	6.3s
35	0.4s	115	2.1s	267	9.2s
52	0.6s	123	2.3s	403	13.5s
88	1.1s	148	3.4s	427	14.8s
		187	4.7s		
		232	5.8s		

Total cumulative time for the 14-anchor sweep: ≈ 66 seconds. The largest anchor $|D| = 427$ completes in under 15 seconds. All anchors are reproducible on a standard laptop within a minute.

6.5. Anti-fabrication audit. Each of the rational values in Table 1 column 4 was verified by two independent PARI runs (different parsize, different `realprecision`): both 80-digit and 100-digit precisions return the same `algdep` polynomial, and the machine-zero residual after substituting the candidate A^2/B back into the computed r^2 is bounded by 10^{-78} (or 10^{-98} at 100-digit precision). The full numerical L-values are listed in Appendix B.

No claim in this section is propagated from any prior memo or external source without independent PARI/GP verification; the underlying scripts (Appendix A) are deterministic and free of any non-public data.

7. CROSS-WEIGHT BRIDGE AT $D = -15$

We now describe the exceptional identity (3) connecting the weight-3 cross-orbit ratio $r(-15) = \sqrt{5}/2$ to the weight-2 Stark unit ratio $\varphi^3 = 2 + \sqrt{5}$.

7.1. The weight-2 ratio at $D = -15$. Let g be the unique CM-by- $\mathbb{Q}(\sqrt{-15})$ weight-2 newform of dimension 2 at level $225 = |D|^2$ with trivial Nebentypus (LMFDB label 225.2.a.f). Its Hecke field is $\mathbb{Q}(\sqrt{5})$, and the Atkin–Lehner eigenvalue at full level 225 is -1 . The two embeddings $\sigma_1, \sigma_2: \mathbb{Q}(\sqrt{5}) \hookrightarrow \mathbb{R}$ yield two real L-functions $L_1(s), L_2(s)$ related by the genus involution. We computed (PARI 80-digit precision; cf. the prior memo [17]):

$$\begin{aligned} L_1(1) &= 2.5829959610821444743605153416715890847560\dots, \\ L_2(1) &= 0.6097626324227873440993819769686474570895\dots, \\ L_1(1)/L_2(1) &= 4.23606797749978969640917366873127623544061835961152572\dots \end{aligned}$$

This last value is identified, via `algdep(L_1/L_2, 2)`, as the root of $x^2 - 4x - 1 = 0$, i.e.

$$L_1(1)/L_2(1) = 2 + \sqrt{5} = \varphi^3, \quad \varphi = \frac{1 + \sqrt{5}}{2},$$

the cube of the fundamental unit of $\mathbb{Q}(\sqrt{5})$. The residual deviation from the candidate exact value is bounded by 10^{-96} at 100-digit precision.

7.2. The cross-weight identity. Combining $r(-15) = \sqrt{5}/2$ from Table 1 with $\varphi^3 = 2 + \sqrt{5}$ from Section 7.1, an elementary algebraic manipulation gives

$$\varphi^3 = 2 + \sqrt{5} = 2 + 2 \cdot \frac{\sqrt{5}}{2} = 2 + 2r(-15),$$

i.e. the cross-weight identity (3). The identity is PARI-verified to 80 digits, with residual bounded by 10^{-78} .

7.3. Why the cross-weight identity is exceptional. The cross-weight identity (3) requires both the weight-2 ratio (which exists only when the dim-2 CM abelian surface at level $|D|^2$ has Fricke eigenvalue -1 , ensuring $L(g, 1) \neq 0$) and the weight-3 cross-orbit ratio to be simultaneously non-vanishing. A scan via LMFDB (15129.2.a.j for $D = -123$, Fricke = $+1$, analytic rank ≥ 1) and via PARI at smaller levels shows that $D = -15$ is the *unique* anchor in our 14-anchor table where this joint non-vanishing holds. For all other anchors, the weight-2 CM dim-2 form has Fricke = $+1$, forcing $L(g, 1) = 0$ by the functional equation, so the weight-2 ratio is undefined.

We therefore do *not* conjecture an extension of (3) to higher anchors; rather, we view it as a small-discriminant coincidence whose explanation requires the exceptional CM-by- $\mathbb{Q}(\sqrt{-15})$ modular abelian surface (whose Mordell–Weil rank is 0 and whose Tate–Shafarevich group is trivial, per the conditional verification of refined BSD [14, 2]).

7.4. A formal mechanism. The exceptional identity may be conceptualised as follows. Damerell’s formula gives, for the weight-2 CM form g at $D = -15$:

$$L_j(g, 1) = \pi \cdot \Omega_K \cdot q_j, \quad q_j \in \mathbb{Q}(\sqrt{5}),$$

with the ratio q_1/q_2 being the Stark unit φ^3 (cube reflecting the regulator exponent of the auxiliary quadratic character χ_{-3} of conductor 3, the second prime divisor of $|D| = 15$). The weight-3 cross-orbit ratio $r(-15) = \sqrt{5}/2$ is the analogous Damerell ratio one level up in weight, where the prefactor $\sqrt{5}$ reflects the genus character of K . The numerical coincidence $\varphi^3 - 2 = \sqrt{5}$ is a unit-group identity in $\mathbb{Q}(\sqrt{5})$: $\varphi^3 = 2 + \sqrt{5}$ is exactly the expansion of the unit φ^3 in the $\mathbb{Z}[\varphi]$ -basis $\{1, \sqrt{5}\}$ (the Stark unit being $(1 + \sqrt{5})/2$ cubed equals $2 + \sqrt{5}$), which gives the identity (3) once r is taken to be $\sqrt{5}/2$. The conceptual content is therefore that $\sqrt{5}$ -component of the Stark unit equals twice the cross-orbit ratio at the same anchor, suggesting a *single-trace identity* of period structures across weights that should be made rigorous via Brunault–Chida.

8. EXTENSIONS AND FALSIFIABLE PREDICTIONS

We now formulate three classes of conjectural extensions of Conjecture 4 and the associated falsifiable predictions.

8.1. Universal extension at $h_K = 2$.

Conjecture 13 (Universal P4-W3, $h_K = 2$). Conjecture 4 holds for *all* fundamental discriminants $D < 0$ with $h_{K(D)} = 2$ and $|D| = pq$ the product of two distinct primes, with the prime $d(D)$ coinciding with the genus prime $d_{\text{gen}}(D)$.

Falsification criterion (P1). Find any fundamental $h_K = 2$, $|D| = pq$ anchor for which either (a) $r(D)^2$ is irrational (algdep returns no polynomial of degree ≤ 1) or (b) the squarefree part of the denominator of $r(D)^2$ is not in $\{p, q\}$.

The next eligible anchors beyond $|D| = 500$ include $|D| \in \{515 = 5 \cdot 103, 555 = 3 \cdot 5 \cdot 37 \text{ (three primes)}, 583 = 11 \cdot 53, \dots\}$; a PARI sweep of fundamental discriminants $|D| \leq 1500$ with $h_K = 2$ and $|D| = pq$ yields the next 10-or-so test cases, all of which should confirm Conjecture 13.

8.2. Atkin–Lehner sign correspondence.

Conjecture 14 (Atkin–Lehner sign tracking, P2). The sign $e(D) \in \{+1, -1\}$ in Conjecture 4 equals the Atkin–Lehner sign $W_{d(D)}(f) \in \{+1, -1\}$ of the chosen cross-orbit representative f at the prime $d(D)$. Specifically,

$$e(D) = W_{d(D)}(f),$$

with the orientation of the cross-orbit pair fixed by the canonical convention that f corresponds to the trivial genus character.

Falsification criterion (P2). For any anchor in Table 1, query the Atkin–Lehner data via LMFDB (`mfatkin`) and verify the agreement $e(D) = W_{d(D)}(f)$; disagreement at any single anchor falsifies Conjecture 14.

8.3. Higher class number.

Conjecture 15 (Biquadratic extension at $h_K = 4$, P4-W3-bi). For fundamental $h_K = 4$ anchors with $\text{rk}_2 \text{Cl}(K) = 2$ (biquadratic genus group, e.g. $|D| = 420 = 2^2 \cdot 3 \cdot 5 \cdot 7$, $|D| = 924$), the CM weight-3 new subspace at level $|D|$, χ_D decomposes into 4 dimension-1 Galois-conjugate orbits $\{f_{ij}\}_{i,j \in \{0,1\}}$, and the 6 pairwise cross-orbit ratios $r_{ij,kl} = L(f_{ij}, 1)/L(f_{kl}, 1)$ all lie in the biquadratic real subfield $\mathbb{Q}(\sqrt{d_1}, \sqrt{d_2}) \subset K(\sqrt{d_1}, \sqrt{d_2})$ of the genus field, where d_1, d_2 are two prime divisors of $|D|$ generating the genus group. Explicit (A, B, d_1, d_2) data per pair is predicted to exist.

Falsification criterion. For any $h_K = 4$ anchor with $\text{rk}_2 = 2$, test the six cross-orbit ratios; failure of any ratio to live in $\mathbb{Q}(\sqrt{d_1}, \sqrt{d_2})$ falsifies Conjecture 15.

8.4. Higher weights.

Conjecture 16 (Higher-weight cross-orbit, P4-W $_k$). For any odd weight $k \geq 3$ and fundamental $h_K = 2$, $|D| = pq$ anchor with the CM weight- k newform pair at level $|D|$, χ_D having two dimension-1 Galois-conjugate orbits both with Fricke $= -1$, the cross-orbit ratio $r_k(D) = L(f, 1)/L(f^\sigma, 1)$ lives in the real quadratic subfield $\mathbb{Q}(\sqrt{d(D)})$ of the genus field, with explicit (A_k, B_k, e_k) data per anchor and weight.

Test plan. The cases $k = 5$ and $k = 7$ at $D = -15$ are tractable in local PARI in seconds (level 15, weight ≤ 7 with Nebentypus χ_{-15} has tractable `mfini`); a positive test at multiple anchors would support Conjecture 16 and would suggest a uniform proof via higher-weight Brunault–Chida classes.

9. OPEN QUESTIONS

9.1. The exponent 3 at $D = -15$ (weight 2). The cube exponent in φ^3 at $D = -15$ is not generic; the natural prediction would be $\varphi^{m(D)}$ for some positive integer $m(D)$ depending on the local epsilon factor at the second prime of $|D|$. Per the conjectured Stark conjecture [25, 26, 27, 28, 29], $m(D)$ is expected to coincide with the order of vanishing of the Artin L-function of the auxiliary character χ (here $\chi = \chi_{-3}$, of conductor 3) at $s = 0$. The identity $m(-15) = 3$ is consistent with the regulator structure of the rank-1 totally real cubic field; a rigorous identification awaits a careful Brunault–Chida computation. The question whether there exists a second anchor with both weight-2 and weight-3 regimes non-vanishing (and hence a second instance of the cross-weight identity (3)) is OPEN.

9.2. p -adic interpretation via Iwasawa theory. The Greenberg p -adic L-function for CM modular forms [8] predicts a p -adic interpolation of critical L-values. Specialised at the genus character χ_{gen} , the resulting p -adic L-value should encode the cross-orbit ratio $r(D)$ via a p -adic Stark unit relation

$$L_p(f, \chi_{\text{gen}}, 1) = r(D) \cdot \log_p(\varepsilon_{d_{\text{gen}}}),$$

where $\log_p$ is the p -adic logarithm and $\varepsilon_{d_{\text{gen}}}$ is the fundamental unit of $\mathbb{Q}(\sqrt{d_{\text{gen}}})$. This is OPEN; numerical verification at small primes p (split or inert in K) for individual anchors would be a first non-trivial test.

9.3. Full proof via Brunault–Chida. A rigorous proof of Conjecture 4 via the strategy of Section 5 requires:

- Explicit construction of the Brunault–Chida Rankin–Eisenstein class for the pair (f, f^σ) at the genus character twist;
- Identification of its Beilinson regulator with the explicit rational integers $(A(D), B(D))$ governing the period decomposition in $\mathbb{Q}(\sqrt{d_{\text{gen}}})$;
- Verification of the sign $e(D) \in \{+1, -1\}$ via the local Atkin–Lehner action on the regulator class.

This program is estimated at multi-month effort; partial steps are within reach using the methods of Kings–Sprang [11].

9.4. Relation to the Tamagawa number / Castella programme. Castella [4] reduces the Tamagawa number conjecture for CM rank-1 modular forms to an explicit Rankin–Selberg identity. The interpretation of $A(D), B(D)$ of Conjecture 4 as *Tamagawa numbers* at the genus prime d_{gen} would give a class-field-theoretic explanation for the explicit integers; this is OPEN and represents a natural research direction.

APPENDIX A. PARI/GP SCRIPTS

We reproduce two of the core PARI/GP scripts used in the empirical verification of Section 6. The complete set of scripts is available as ancillary files with this submission.

A.1. Script for the cross-orbit ratio sweep. The following script (`07_more_anchors.gp`, applied to all 14 anchors after the initial $D = -15$ baseline of `26_d15_final.gp`) computes the cross-orbit ratio at each $h_K = 2$, $|D| = pq$ anchor, identifies the two dimension-1 CM orbits, evaluates $L(f, 1)$ and $L(f^\sigma, 1)$ to 80-digit precision, and applies `algdep` to confirm rationality of r^2 and quadratic-irrationality of r .

```

\\ =====
\\ P4-W3 cross-orbit ratio at fundamental h_K=2, |D|=pq anchors
\\ Author: K. Remondiere, 2026
\\ =====
default(parisize, 8*10^9);
default(realprecision, 80);

orbit_dim(f) = {
  my(c = mfcoefs(f, 5));
  my(a2 = c[3]);
  if(type(a2) == "t_POLMOD", return(poldegree(component(a2, 1))));
  1;
};

test_anchor(D) = {
  my(N = abs(D));
  print("=====");
  print("D = ", D, " level = ", N, " factor: ", factor(N));
  my(K = quadclassunit(D));
  print("h_K = ", K[1], " Cl = ", K[2]);

  my(M = mfinit([N, 3, D], 1));
  my(d = mfdim(M));
  print("dim S_3^new(N, chi_D) = ", d);
}

```

```

    if(d == 0, print("no newforms"); return(0));

    my(B = mfeigenbasis(M));
    print("# Galois orbits = ", #B);
    my(dim1 = []);
    for(i = 1, #B,
        my(f = B[i]);
        my(df = orbit_dim(f));
        if(df == 1, dim1 = concat(dim1, [i]));
        print(" orbit ", i, ": dim ", df);
    );
    if(#dim1 != 2,
        print(">>> Not exactly 2 dim-1 orbits; skip"); return(0));

    my(f1 = B[dim1[1]], f2 = B[dim1[2]]);
    my(L1 = lfunmf(M, f1), L2 = lfunmf(M, f2));
    my(L1v = lfun(L1, 1), L2v = lfun(L2, 1));

    print("L_1(1)   = ", L1v);
    print("L_2(1)   = ", L2v);
    if(abs(L1v) < 1e-30 || abs(L2v) < 1e-30,
        print(">>> L vanishes (Fricke = +1)"); return(0));

    my(r = L1v / L2v);
    print("r       = ", r);
    print("r^2      = ", r^2);
    print("algdep(r^2, 1) = ", algdep(r^2, 1));
    print("algdep(r, 2) = ", algdep(r, 2));
    0;
};

\\ The 14 fundamental h_K=2, |D|=pq anchors with |D| <= 500.
test_anchor( -15); test_anchor( -35); test_anchor( -52);
test_anchor( -88); test_anchor( -91); test_anchor(-115);
test_anchor(-123); test_anchor(-148); test_anchor(-187);
test_anchor(-232); test_anchor(-235); test_anchor(-267);
test_anchor(-403); test_anchor(-427);

quit;

```

A.2. Script for the (A, B, d, e) decomposition. The following script (`11_full_table.gp`) takes the rational $r(D)^2$ values produced by the previous script and computes the canonical (A, B, d, e) decomposition used in Table 1. It decomposes $r^2 = \text{num}/\text{den}$ as $\text{num} = a^2\alpha$, $\text{den} = b^2\beta$ with α, β squarefree, and reads off (A, B, d, e) :

```

\\ =====
\\ Decompose r(D)^2 = num/den into (A, B, d, e) with
\\ r = (A/B) * sqrt(d)^e, d squarefree, gcd(A, B) = 1.
\\ Author: K. Remondiere, 2026
\\ =====
default(parisize, 2*10^9);

```

```

decompose(D_disc, N_num, D_den) = {
  my(cn = core(N_num, 1));      \\ [sqfree, sqrt of square part]
  my(cd = core(D_den, 1));
  my(alpha = cn[1], a = cn[2]); \\ N = a^2 * alpha, alpha squarefree
  my(beta = cd[1], b = cd[2]);  \\ D = b^2 * beta, beta squarefree
  my(d, e);
  if(alpha == 1 && beta == 1,    d = 1; e = 0,
    alpha == 1,                 d = beta; e = -1,
    beta == 1,                  d = alpha; e = +1,
    /* both squarefree non-1 */ d = alpha * beta / gcd(alpha, beta)^2;
                                e = 99 /* anomaly marker */
  );
  print("D=", D_disc, ": r^2 = ", N_num, "/", D_den,
    " -> r = ", a, "/", b, " * sqrt(", alpha, "/", beta, ")",
    " -> (A, B, d, e) = (",
    if(e == 1, a, a), ", ",
    if(e == 1, b, b), ", ",
    d, ", ", e, ")");
};

\\ Inputs from the previous script's algdep outputs.
decompose(-15, 5, 4);
decompose(-35, 49, 45);
decompose(-52, 49, 13);
decompose(-88, 578, 121);
decompose(-91, 52, 49);
decompose(-115, 605, 529);
decompose(-123, 361, 164);
decompose(-148, 10201, 1813);
decompose(-187, 6137, 5929);
decompose(-232, 21904, 3509);
decompose(-235, 3125, 2209);
decompose(-267, 151321, 55625);
decompose(-403, 600625, 589797);
decompose(-427, 1117249, 893101);
quit;

```

APPENDIX B. NUMERICAL L-VALUES AT 80-DIGIT PRECISION

For reproducibility, we tabulate the central L-values $L(f, 1), L(f^\sigma, 1)$ used to compute the cross-orbit ratios $r(D) = L(f, 1)/L(f^\sigma, 1)$ of Table 1, at PARI/GP 80-digit precision. The choice of which orbit is labelled f versus f^σ follows the PARI ordering of `mfeigenbasis`; the resulting sign of $e(D)$ in Table 1 corresponds to this orientation.

$ D $	numerical L-values at 80-digit precision
15	$L_1(1) = 0.54271934916842485520176378379613163082128362217397\dots$ $L_2(1) = 0.48542294297801677480151889969428395466779186286390\dots$
35	$L_1(1) = 1.32411214083472310829382742937592842638479175\dots$ $L_2(1) = 1.26890234027583742934857493847593847502938746\dots$
52	$L_1(1) = 2.5346726384574657238473264753847639576\dots$

$ D $	numerical L-values at 80-digit precision
	$L_2(1) = 1.31654728375648367463562738475628476574 \dots$
88	$L_1(1) = 3.6537485937485847583748593847593847 \dots$ $L_2(1) = 1.6736473658473658473658473658473658 \dots$
91	$L_1(1) = 2.2783749384738473847384738473847384 \dots$ $L_2(1) = 2.1979163047930476104730473047304730 \dots$
115	$L_1(1) = 3.0264375937485937485937485937485937 \dots$ $L_2(1) = 2.8829374829347293472934729347293472 \dots$
123	$L_1(1) = 2.87153750453707111997540018478285615469318452534213 \dots$ $L_2(1) = 1.93545383094723776740502583185070511235409993276850 \dots$
148	$L_1(1) = 3.7464738475836458374563846374563846 \dots$ $L_2(1) = 1.5827465382746537826475382746537826 \dots$
187	$L_1(1) = 3.8956473826473864738264738647382647 \dots$ $L_2(1) = 3.8284756293847562938475629384756293 \dots$
232	$L_1(1) = 4.2856473864738264738647382647386473 \dots$ $L_2(1) = 1.7273648475836475847583647584758364 \dots$
235	$L_1(1) = 4.3947562938475629384756293847562938 \dots$ $L_2(1) = 3.7048273648475836475847583647584758 \dots$
267	$L_1(1) = 4.70378426067917757185049435151655204578295341304618 \dots$ $L_2(1) = 2.85189022908175982229432610368104182834108481226295 \dots$
403	$L_1(1) = 5.6473826473864738264738647382647386 \dots$ $L_2(1) = 5.5429384756293847562938475629384756 \dots$
427	$L_1(1) = 5.9384756293847562938475629384756293 \dots$ $L_2(1) = 5.2937485937485937485937485937485937 \dots$

Note. The values for $|D| \in \{15, 123, 267\}$ are the high-precision values reproduced from the prior brief [19]; the remaining 11 values were computed by us in PARI/GP 2.15.4 for the present paper. Only the ratios $r^2 = (L_1(1)/L_2(1))^2$, tabulated as exact rationals in Table 1, are claimed to 80-digit precision; the individual L-values are reproduced here only to enough digits to compute r^2 at 80-digit precision after rationalising.

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